JNTUH USED 28-04-18AM Set-1

R13

Code No: 118BA

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, April - 2018
DATABASE SECURITY
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A

(25 Marks)

1.a)	What are non fraudulent attacks?	[2]
b)	What is inference channel? How is it useful in database security?	[3]
c)	What are the access modes of Biba security model?	[2]
d)	Write about explicit flow in lattice security model.	[3]
e)	What is statistical filter?	[2]
f)	Write the advantages of kernel based secure operating system.	[3]
g)	Write the basic difference between ORION and SORION model.	[2]
h)	What is Role lattice?	[3]
i)	What are the entities of message filter?	[2]
j)	Compare Implicit and explicit authorization.	[3]

PART - B

(50 Marks)

2.a)	What are the requirements of database protection? Explain it in detail.	
b)	With a neat sketch, explain elements of security model of security system.	[5+5]
	OR	
3.a)	Explain, how flow control is useful in control security.	
b)	Give a detailed note on access matrix control model.	[5+5]
4.a)	Elaborate axioms of Dion secure model.	
b)	Discuss about flow control mechanisms.	[5+5]
	OR	
5.a)	Describe TCB secure model.	
b)	Explain different user authentication methods which are useful providing security to database.	[5+5]
6.a)	Discuss about various security mechanisms in DBMS.	
b)	Write the guidelines for mechanism selection of secure database.	[5+5]
	OR	
7.a)	Elaborate conceptual techniques of inference protection.	
b)	With a neat sketch, Explain structure of IDES.	[5+5]

8.a) Explain various axioms of SORION.
b) Explain the entities of SORION model with a neat sketch. [5+5]

OR

9.a) Write the security policies for frame based model.
b) Discuss about the elements of object oriented database. [5+5]

10.a) Discuss about access modes of ORION model.
b) Write the elements of active database. [5+5]

OR

11.a) Write message filtering algorithm.
b) Discuss access control and authorization techniques in ORION model. [5+5]

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